The completeness theorem

The Open Logic Project contributors

openlogic:first-order-logic/completeness:local-snapshot-20261013;
CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two lem:truth citations target the two sibling instances of the shared label (no-identity term model at source line 1312, identity quotient at source line 1876); the navigation stub is anchored at the first instance and this note records the second

Editorial note. The label thm:completeness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Henkin saturation extension. Same unit, \ollabel{lem:henkin} (source line 1559). FOL branch: extends Gamma to a saturated consistent superset Gamma'.
- Lindenbaum extension. Same unit, \ollabel{lem:lindenbaum} (registered card). Extends Gamma' (or Gamma directly) to a complete consistent Gamma*.
- Truth Lemma for the term model. Same unit, \ollabel{lem:truth} (source line 1312). No-identity case: M(Gamma*) satisfies exactly the sentences in Gamma*.
- Truth Lemma for the quotient model. Same unit, \ollabel{lem:truth} (source line 1876). Identity case: the quotient of the term model by the approx relation satisfies exactly the sentences in Gamma*.

Source reference. Original lemma lem-henkin: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1559–1559. Prerequisite pointer invoked at proof line(s) 1010; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-truth: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1312–1312. Prerequisite pointer invoked at proof line(s) 1022; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-lindenbaum: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 2006–2006. Prerequisite pointer invoked at proof line(s) 1011; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Completeness Theorem)**.** *Let* Γ *be a set of sentences. If* Γ *is consistent, it is satisfiable.*

Proof. Suppose Γ is consistent. By lem-henkin, there is a saturated consistent set $\Gamma' \supseteq \Gamma$.By lem-lindenbaum, there is a $\Gamma^* \supseteq \Gamma'$ which is consistent and complete. Since $\Gamma' \subseteq \Gamma^*$, for each formula $A(x)$, Γ^* contains sentence of the form $\exists x\, A(x) \to A(c)$and so Γ^* is saturated. If Γ does not contain $=$, then bylem-truth, $\mathfrak{M}(\Gamma^*) \vDash$ iff $A \in \Gamma^*$. From this it follows in particular that for all $A \in \Gamma$, $\mathfrak{M}(\Gamma^*) \vDash$, so Γ is satisfiable. If Γ does contain $=$, then by lem-truth, for all sentences A, $[M]{\approx} \vDash A$ iff $A \in \Gamma^*$. In particular, $[M]{\approx} \vDash A$ for all $A \in \Gamma$, so Γ is satisfiable. $\square$